Computational Biology Subsidary Notes

Lecture 17: Dynamical Systems

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1 Keywords

- ODEs, nullclines, fixed points, stability analysis, bifications

2 Main Points

2.1 Understanding ODEs

- Understanding systems of ODEs is a classic problem in what is known as *dynamical systems*
- To save on notation I'm going to drop writing concentrations of X as $[X]$ (this is just me being lazy)
- I am going to use examples to illustrate dynamics rather concentrate on biology, although you can try to imagine a scenario where the dynamics describes some biological process
- The type of equations I am considering are a set of equations for variables (molecular species) $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$ governed by

$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X})$$

  - $f_i(\boldsymbol{X})$ describes a set of terms that potentially depends on the concentrations all the other species

2.1.1 Two Species

- We consider a system where X_2 acts as a repressor for the production of X_1 and X_1 turns into X_2

$$\varnothing \xrightarrow{f(X_2)} X_1 \qquad X_1 \xrightarrow{k_3} \varnothing \qquad X_1 \xrightarrow{k_5} X_2$$
$$\varnothing \xrightarrow{k_2} X_2 \qquad X_2 \xrightarrow{k_4} \varnothing$$

  with

$$f(X_2) = \frac{k_1}{1 + (X_2/K)^n}$$

- The ODE describing these species are

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K)^n} - k_3\, X_1(t) - k_5\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_2 + k_5\, X_1(t) - k_4\, X_2(t)$$

- To solve the ODE we need to provide some initial conditions

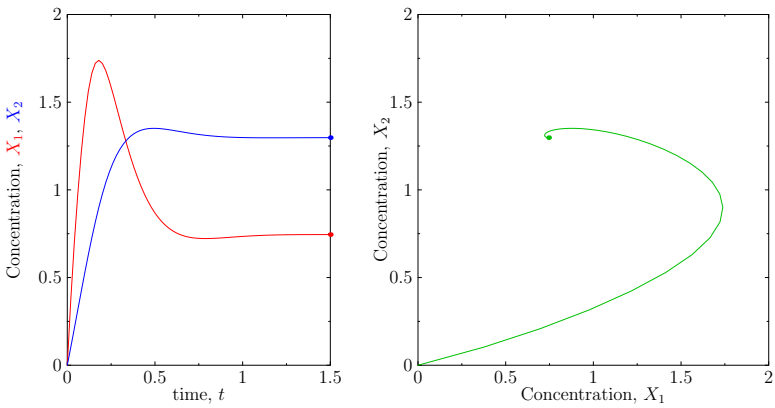

Figure 1: Dynamics for two interaction species

- In Figure 1 we show the dynamics assuming $X_1 = X_2 = 0$ and with parameter values of $K = 1$, $k_1 = 20$, $k_2 = k_3 = k_4 = 5$, $k_5 = 2$ and $n = 4$
- However, it is useful to understand the dynamics from any initial conditions
- In Figure 2 we show some trajectories starting from different initial points as well as the vector field of derivatives (flow fields shown as arrows)

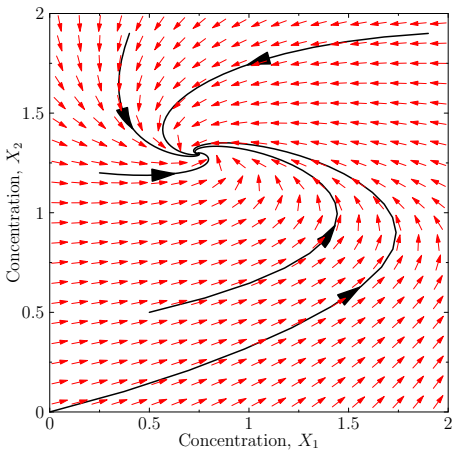

Figure 2: Trajectory and flow field for two species system

- To get a better analytic understanding of what is going on we can compute the **nullclines**: these are curves where $\frac{\mathrm{d}X_i}{\mathrm{d}t} = 0$
- For this system

$$\frac{\mathrm{d}X_1}{\mathrm{d}t} = \frac{k_1}{1 + (X_2/K)^n} - k_3\,X_1 - k_5\,X_1 = 0 \qquad \Rightarrow \qquad X_1 = \frac{k_1}{(k_3 + k_5)(1 + (X_2/K)^n)}$$

$$\frac{\mathrm{d}X_2}{\mathrm{d}t} = k_2 + k_5\,X_1 - k_4\,X_2 = 0 \qquad \Rightarrow \qquad X_2 = \frac{k_2 + k_5\,X_1}{k_2}$$

- Where the nullclines intersect we have a *stationary point*
- We plot the nullclines in Figure 3

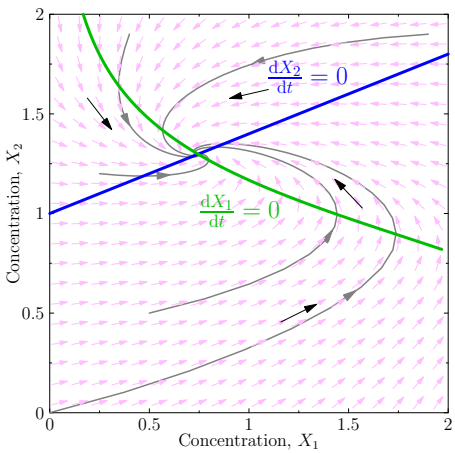

Figure 3: Nullclines for two species system

2.2 Stability

- Having identified fixed points (where nullclines intersect) we investigate their stability
- Let us consider a slightly different system
- Here X_1 and X_2 are proteins that inhibit each other's expression

$$\frac{dX_1(t)}{dt} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{dX_2(t)}{dt} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- If we consider the symmetric case $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 4$ (later we explore changing the parameters)
- We can again compute the flow fields and nullclines
- These are shown in Figure 4

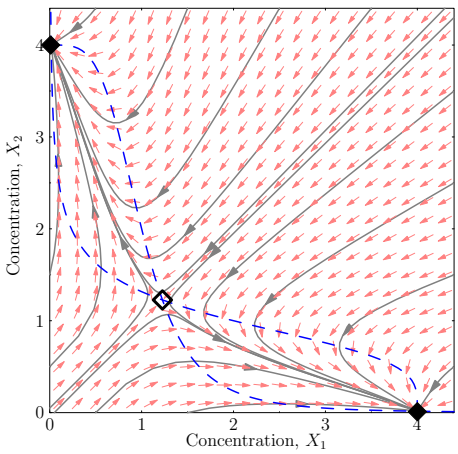

Figure 4: Nullclines and flow field for bistable system

- The nullclines intersect at three points leading to three fixed-point
- The flow fields show that almost all trajectories flow either to the fixed point with $X_1 \gg X_2$ or with $X_2 \gg X_1$

- There is a third unstable fixed point when $X_1 = X_2$ but this is unstable to any perturbation that breaks the symmetry
- Although it seems that we can investigate fixed-points graphically, this only works in low dimensions (where we have a few species) and doesn't give us insight on how the stability depends on the parameters of the model

2.2.1 Fixed-Point Analysis

- As I said previously these systems are highly non-linear and it is very unlikely that we can solve the dynamics in closed form
- However, there is a very general procedure for understanding the stability of fixed-points
- We linearise the dynamics around the fixed point
- Recall that we have a set of ODEs
$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X})$$
- Now assume we are at a point $\boldsymbol{X} = \boldsymbol{X}^* + \boldsymbol{\epsilon}$ where $\boldsymbol{X}^*$ is a fixed point
- We can Taylor expand $f_i(\boldsymbol{X})$ around $\boldsymbol{X}^*$
$$\frac{\mathrm{d}X_i}{\mathrm{d}t} = f_i(\boldsymbol{X}^* + \boldsymbol{\epsilon}) \approx f_i(\boldsymbol{X}^*) + \boldsymbol{\epsilon}^\mathsf{T}\boldsymbol{\nabla} f_i(\boldsymbol{X}^*) + O(\boldsymbol{\epsilon}^2) = \boldsymbol{\epsilon}^\mathsf{T}\boldsymbol{\nabla} f_i(\boldsymbol{X}^*) + O(\boldsymbol{\epsilon}^2)$$
where we have used $f_i(\boldsymbol{X}^*) = 0$ as we are at a fixed point
- Writing this in vector form
$$\frac{\mathrm{d}\boldsymbol{X}}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{X}^* + \boldsymbol{\epsilon}) = \mathbf{J}\,\boldsymbol{\epsilon} + O(\boldsymbol{\epsilon}^2)$$
  - where $\mathbf{J}$ is the **Jacobian** matrix with elements
$$J_{ij} = \left.\frac{\partial f_i(\boldsymbol{X})}{\partial X_j}\right|_{\boldsymbol{X}=\boldsymbol{X}^*}$$
- Since $\boldsymbol{\epsilon}(t) = \boldsymbol{X}(t) - \boldsymbol{X}^*$ then
$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{X}(t)}{\mathrm{d}t} \approx \mathbf{J}\,\boldsymbol{\epsilon}(t)$$
- To solve this we use the eigenvalue decomposition of $\mathbf{J}$
$$\mathbf{J} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}$$
  - Where $\boldsymbol{\Lambda}$ is a diagonal matrix whose elements $\Lambda_{ii} = \lambda_i$ are eigenvalues of $\mathbf{J}$ and the columns of $\mathbf{V}$ are the eigenvectors
  - For non-symmetric matrices (such as $\mathbf{J}$ usually) the eigenvalues are either real or appear in pairs, λ_i and λ_{i+1} where $\lambda_{i+1} = \lambda_i^*$ (i.e. they are complex conjugate pairs
  - In this case the corresponding eigenvectors $\boldsymbol{v}_i$ and $\boldsymbol{v}_{i+1}$ will also occur in complex conjugate pairs $\boldsymbol{v}_{i+1} = \boldsymbol{v}_i^*$
  - Thus
$$\frac{\mathrm{d}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} = \mathbf{V}\boldsymbol{\Lambda}\,\mathbf{V}^{-1}\,\boldsymbol{\epsilon}(t) \qquad \Rightarrow \qquad \frac{\mathrm{d}\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)}{\mathrm{d}t} \approx \boldsymbol{\Lambda}\,\mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$$
    where we multiply both sides by $\mathbf{V}^{-1}$

- Defining $\hat{\boldsymbol{\epsilon}}(t) = \mathbf{V}^{-1}\boldsymbol{\epsilon}(t)$ then

$$\frac{\mathrm{d}\hat{\boldsymbol{\epsilon}}(t)}{\mathrm{d}t} \approx \boldsymbol{\Lambda}\,\hat{\boldsymbol{\epsilon}}(t) \qquad \text{or} \qquad \frac{\mathrm{d}\hat{\epsilon}_i(t)}{\mathrm{d}t} \approx \lambda_i\,\hat{\epsilon}_i(t)$$

- This aligns the perturbations with the eigenvectors of $\mathbf{J}$
- We can solve these equations giving

$$\hat{\epsilon}_i(t) \approx \mathrm{e}^{\lambda_i\,t}\;\hat{\epsilon}_i(0)$$

  * If λ_i is real and positive the any perturbation in the direction $\boldsymbol{v}_i$ will grow with time
  * If λ_i is real and negative then any perturbation in the direction $\boldsymbol{v}_i$ will shrink
  * If λ_i and λ_{i+1} are complex conjugate pairs then the trajectory in the plane spanned by $\boldsymbol{v}_i$ and $\boldsymbol{v}_{i+1}$ will be a spiral around the fixed-point
    · It will spiral out if the real part of the eigenvalues are positive and spiral in if the real values of the eigenvalues are negative
- The fixed point is stable if **all** eigenvalues satisfy $\Re\,\lambda_i < 0$

- For the bistable system, the unstable fixed point is at $X_1 = X_2 = 1.226$
- The Jacobian is

$$\mathbf{J} = \begin{pmatrix} -5 & -13.87 \\ -13.87 & -5 \end{pmatrix}$$

  This has eigenvalues are 8.868 and -18.868 in the directions $\binom{1}{-1}$ and $\binom{1}{1}$, respectively. Clearly this is unstable to perturbations that break the symmetry between X_1 and X_2.

- **Oscillator**
  - We consider the process of generating X_2 through an intermediary X_1 where the production of X_2 from X_1 is catalysed by the presence of X_2

$$\varnothing \xrightarrow{k_0} X_1 \qquad X_1 \xrightarrow{f(X_2)} X_2 \qquad X_2 \xrightarrow{k_2} \varnothing$$

    where $f(X_2) = k_1\,(1 + (X_2/K)^n)$
  - This leads to the coupled pair of ODEs

$$\begin{aligned} \frac{\mathrm{d}X_1(t)}{\mathrm{d}t} &= k_0 - k_1\,\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right)\,X_1(t) \\ \frac{\mathrm{d}X_2(t)}{\mathrm{d}t} &= k_1\,\left(1 + \left(\frac{X_2(t)}{K}\right)^n\right)\,X_1(t) - k_2\,X_2(t) \end{aligned}$$

  - In Figure 10 we show the dynamics for $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$ and $n = 2$
  - The fixed-point is given by

$$\boldsymbol{X}^* = \left(\frac{k_0}{k_1(1 + (k_0/(K\,k_2))^n)}, \frac{k_0}{k_2}\right) = \left(\frac{200}{89}, \frac{8}{5}\right)$$

  - The Jacobian is given by

$$\mathbf{J} = \begin{pmatrix} -k_1\,\left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & -\frac{n\,k_1}{K}\,\left(\frac{X_2^*}{K}\right)^{n-1}\,X_1^* \\ k_1\,\left(1 + \left(\frac{X_2^*}{K}\right)^n\right) & \frac{n\,k_1}{K}\,\left(\frac{X_2^*}{K}\right)^{n-1}\,X_1^* - k_2 \end{pmatrix} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix}$$

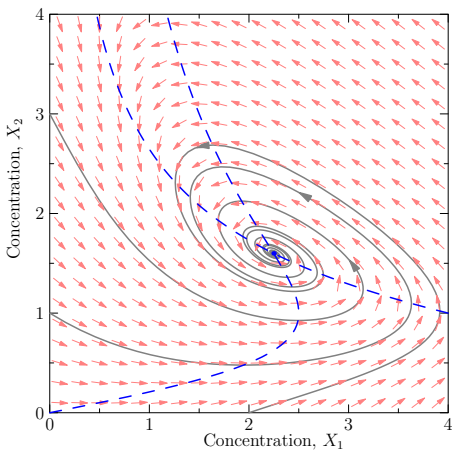

Figure 5: Dynamics of Oscillator

- Plugging this into Matlab (using `eig(J)`)

$$\mathbf{J} = \begin{pmatrix} \frac{-89}{25} & \frac{-640}{89} \\ \frac{89}{25} & \frac{195}{89} \end{pmatrix} = \mathbf{V} \begin{pmatrix} -0.68 + 4.16\mathrm{i} & 0 \\ 0 & -0.68 - 4.16\mathrm{i} \end{pmatrix} \mathbf{V}^{-1}$$

  where

$$\mathbf{V} = \begin{pmatrix} 0.82 & 0.82 \\ -0.33 - 0.47\mathrm{i} & -0.33 + 0.47\mathrm{i} \end{pmatrix}$$

- The eigenvalues are $-0.68 \pm 4.16\mathrm{i}$ with eigenvectors of $\left(\begin{smallmatrix} 0.82 \\ -0.33 \pm 0.47\mathrm{i} \end{smallmatrix}\right)$
- Eigenvalues are a pain to compute by hand, but for 2×2 matrices we can compute them from the trace, $T = \operatorname{tr} \mathbf{J} = J_{11} + J_{22}$, and determinant $D = \det(\mathbf{J}) = J_{11}\, J_{22} - J_{12}\, J_{21}$ where

$$\lambda_{1,2} = T/2 \pm \sqrt{T^2/4 - D}$$

- The system is stable (the real part of the eigenvectors are negative), but the large complex component causes a significant spiralling

2.3 Bification

- In understanding dynamical systems we can study how the fixed points change as we vary the parameters of the system
- Fixed points tend to appear in pairs, one stable and one unstable
- Clearly if we have two stable fixed point then there will be some manifold that divides the basins of attraction of the two fixed points
- On this manifold there will usually exist an unstable fixed-point
- To illustrate this we return to the bistable problem

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = \frac{k_1}{1 + (X_2(t)/K_2)^{n_1}} - k_3\, X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = \frac{k_2}{1 + (X_1(t)/K_1)^{n_2}} - k_4\, X_2(t)$$

- Earlier on we examine the symmetric case when $k_1 = k_2 = 20$, $K_1 = K_2 = 1$, $k_3 = k_4 = 5$, $n_1 = n_2 = 2$

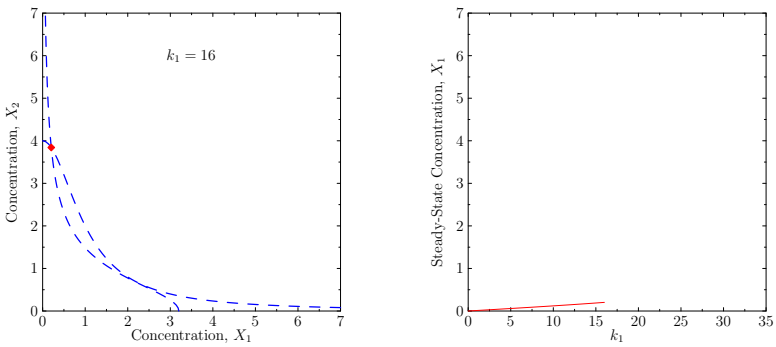

Figure 6: Dynamics of Oscillator

- Now as we vary k_1 from a very small value to large value we change the nullcline associated with $\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = 0$
- For $k_1 < 16$ there is only one crossing of the nullclines leading to one fixed-point
- Again for $k_1 > 29$ there is only one fixed-point which persists as we increase k_1 further

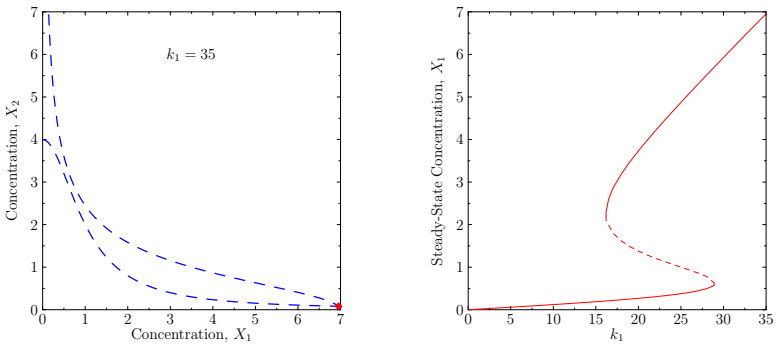

Figure 7: Dynamics of Oscillator

- The appearance of a pair of fixed-point is known as a **bification**
- This type of behaviour can result in a *hysteresis loop* where as you change parameters you make a sudden jump when a stable fixed-point annihilates with an unstable fixed-point. Reversing the change in the parameter you stay attracted to the second fixed point.

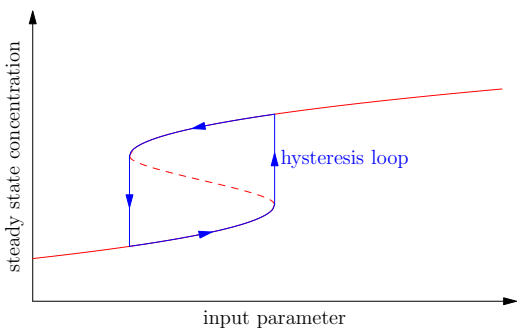

Figure 8: Dynamics of Oscillator

2.3.1 Hopf Bification

- A second mechanism where the nature of the fixed-point changes is when the stability of the fixed point changes

- This happens in the autocatalytic reaction we looked at earlier

$$\frac{\mathrm{d}X_1(t)}{\mathrm{d}t} = k_0 - k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t)$$
$$\frac{\mathrm{d}X_2(t)}{\mathrm{d}t} = k_1 \left(1 + \left(\frac{X_2(t)}{K}\right)^n\right) X_1(t) - k_2\, X_2(t)$$

  with $k_0 = 8$, $k_1 = 1$, $K = 1$, $k_2 = 5$

- We consider what happens as we vary n
  - For small n we have a stable fixed-point

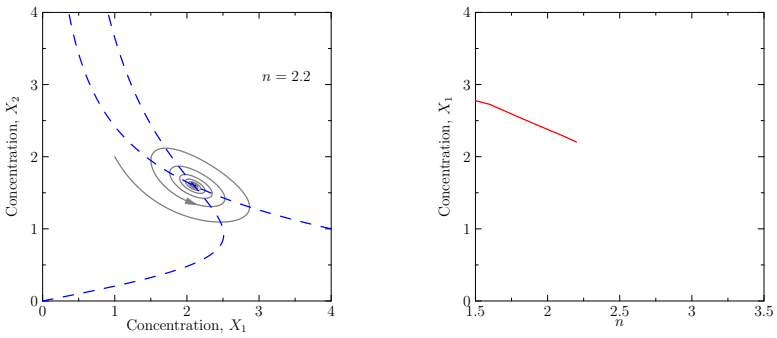

Figure 9: Dynamics of Oscillator

  - As we increase n the fixed point becomes unstable

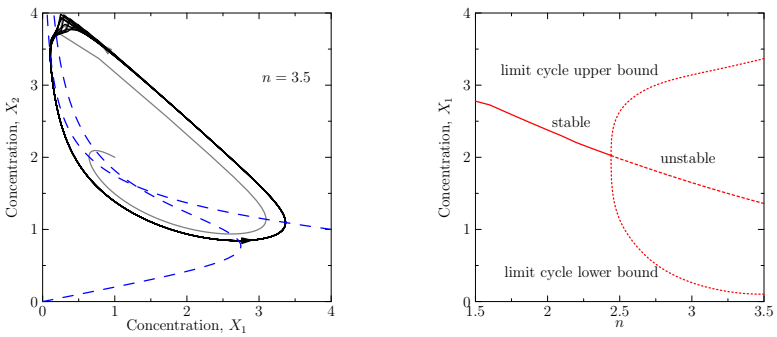

Figure 10: Dynamics of Oscillator

  - In this case the dynamics converges to a *limit cycle* that oscillates around the unstable fixed point

- This is known as a *Hopf bification*